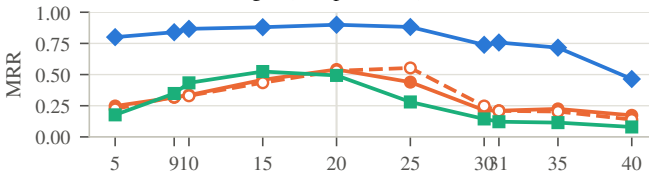
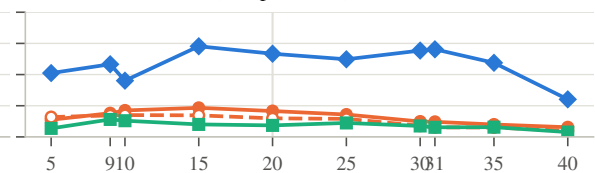
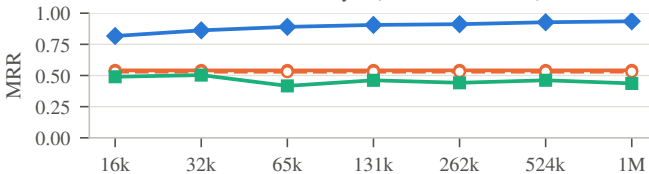
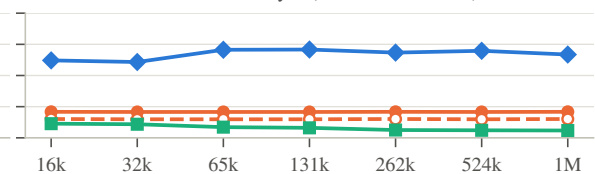


Lean 4 $\rightarrow$ HOL Light (317 problems)Lean 4 $\rightarrow$ Metamath (482 problems)

Gemma-2-9B layer (131k-feature SAE)



Gemma-2-9B layer (131k-feature SAE)



SAE width at layer 20 (features; average L0 48–58)

Dense

Dense, PC1 removed

SAE, unweighted

SAE width at layer 20 (features; average L0 48–58)

SAE-IDF (ours)